# ZEQ OS: Universal Proper-Time Modulation

**Synchronized Physics Across Quantum, Classical, and Relativistic Domains**

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Zeq OS 1.287 Hz Research
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We propose the **ZEQ Equation**, a universal proper-time modulation described by

$$R(t) = S(t)[1 + \alpha \sin(2\pi f t + \varphi_0)]$$

where $S(t)$ represents any standard physical prediction, $\alpha = 1.29 \times 10^{-3}$ is the dimensionless modulation amplitude, and $f = 1.287$ is the modulation frequency. Rather than modifying fundamental physical laws, this framework synchronizes them to a shared computational rhythm.

**Key Insight:** Standard physics is recovered exactly as the time-average over the modulation period $T = 1/f \approx 0.777$ (one zeqond), ensuring full compatibility with all established experimental results.

### Parameters

- $R(t)$: Observable physical quantity
- $S(t)$: Standard physics prediction (Newton, Einstein, Schrödinger)
- $\alpha = 1.29 \times 10^{-3}$: Dimensionless modulation amplitude
- $f = 1.287$ : Universal synchronization frequency
- $\varphi_0$: System-specific phase offset

### Terminology

- **HulyaPulse**: 1.287 Hz system frequency (fundamental clock cycle)
- **Zeqond**: 777 ms timer per pulse (true computational second, $T = 1/f$)

**Corollary:** All dimensionless physical constants remain invariant under this modulation.

This work is intended to be evaluated exclusively through computation and measurement: the equation should be directly applied, cross-correlated across domains, and tested experimentally.

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**Reference:**

Zeq, H., & Zeq, A. (2025). *ZEQ OS - Evolution of Mathematics: A Synchronized Computational Formalism Featuring HULYAS Math, Kinematic Operators and the 1.287 Hz HulyaPulse for Analysis Across Quantum to Relativity. A Theory of Everything?* Zenodo. https://doi.org/10.5281/zenodo.15825138

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